



GARASU · Quick Start

GLASS AGING, RELAXATION AND SIMULATION UTILITY ·
BETA

Version 1.5 · October 2026
Full details: [GARASU User Guide](#)

1 · Open GARASU

In the browser (nothing to install)

wtakeda-research.github.io/GARASU

Desktop app

Download from github.com/wtakeda-research/GARASU/releases/latest.

Mac: open the `.dmg`, drag GARASU into Applications, open it. First time only: **System Settings** → **Privacy & Security** → **Open Anyway**.

Windows: run `GARASU-Setup-... .exe`. If warned: **More info** → **Run anyway**.

The app updates itself: accept **Restart now** when offered.

Learn

experiments and models explained, with live simulations

Lab

quick interactive runs, three models side by side

Fitter

your thermal history and data → compute or fit

Data Analysis

raw differential scanning calorimetry (DSC) heat flow, enthalpy recovery, volume relaxation

2 · Good to know

- **Nothing runs until you click.** In the Fitter the model runs only on **Compute** or **Run fit**; any change clears the old result.
- **Temperature K / °C** and **Time axis** (top right) apply everywhere; K/°C also converts inputs and T_g . Time axis *auto* shows real time whenever the temperature changes (scans), log time for isothermal holds.
- **Try before your own data:** every tool has examples (Fitter **Examples ▼**, Data Analysis **Load example data**).
- **Projects:** **New project** in the header keeps normalized scans, fits and sessions of one study together; **Save to project** in Data Analysis and in the Fitter results.
- **Figures:** **Export ▼** on every plot saves PNG, JPEG, SVG or CSV.
- **Private:** everything runs on your computer; no data are uploaded.
- **Help:** see the User Guide, or report issues at github.com/wtakeda-research/GARASU/issues.

Please cite GARASU (Zenodo, doi:10.5281/zenodo.23091595) and W. Takeda and P. Lucas, *J. Chem. Phys.* **160**, 174504 (2024) and *Mater. Adv.* **7**, 5729 (2026) when GARASU contributes to your work. GARASU is free software under the MIT License (© 2026 Arizona Board of Regents on behalf of the University of Arizona).

3 · Three things to try

A · Simulate a DSC scan after aging (Fitter)

≈ 2 min

- 1 Open **Fitter**. In step 1 **Thermal history**, choose the template **Aging below T_g** , then **DSC heating**.
- 2 Step 3 **Model & parameters**: keep **Takeda–Lucas** or pick a **Preset** (e.g. **Selenium**).
- 3 Step 4 **Compute & fit**: click **Compute**. You get T_f , normalized C_p , relaxation during the hold, τ and, for the Takeda–Lucas (TL) model, the map of local fictive temperatures.

B · Fit a model to data (Fitter)

≈ 3 min

- 1 **Examples ▼** → **Aging + DSC with synthetic enthalpy data** (or add your own data in step 2 with **+ Add dataset**, linked to the matching segment).
- 2 Step 3: tick the parameters to fit and check their min/max.
- 3 Step 4: open **Fit parameters**, choose *Local* (fast) or *Global* (thorough), click **Run fit**. Read the values $\pm$ errors, R^2 , and the correlation matrix.

C · Analyze raw DSC heat flow (Data Analysis)

≈ 3 min

- 1 Open **Data Analysis** → **+ Add run**. Load your file or paste columns; check temperature and heat-flow columns, units, the **sign convention** (endo up/down), rate and mass. For scans after annealing, enter t_a and T_a .
- 2 Set the **glass range** and **liquid range** (or **Auto ranges**). Read T_f' , the midpoint and ΔC_p ; **Apply to all runs**.
- 3 **Tab 2 Enthalpy recovery**: choose the unaged reference and enter T_a to get ΔH , T_f and $\phi(t_a)$ with a Kohlrausch–Williams–Watts (KWW) fit. **Tab 3** fits volume (density, length...) during annealing.

